REFERENCE TRANSLATION prepared by Van Phong Technical and Trading Co., Ltd. The published original is the authoritative text; see it for figures.

On Crack Propagation in Rock upon Impact

V. I. Bolobov, B. Le-Thanh, V. A. Plashchinsky

Obogashchenie Rud (Mineral Processing Journal), 2019, No. 6, pp. 3–8 · UDC 622.732 : 539.422 · DOI: 10.17580/or.2019.06.01

Abstract

Based on the assumption that the condition for the destruction of a rock fragment upon impact is that the emerging crack reaches one of the free surfaces of the fragment, the propagation velocity of the fracture crack v_{cr} was estimated experimentally and by calculation on samples of granite from the Vyborg massif. Using an experimental setup, the most effective distance h = l_{eff} of the impact point from the edge of the lateral surface of granite fragments was determined, at which the impact causes a piece of rock of maximum volume to break off. The dependence of this value on the diameter d of the blunting flat of the striker was established, and the length of the fracture crack l_{cr} was determined. The parameters of a single impact were calculated — the rock resistance force N_m, the contact stress σ_c, the penetration α_p of the striker into the rock and the impact duration t_m — and their dependence on d was established. Taking the calculated t_m into account, the velocity v_{cr} was determined. Regardless of the values of d and t_m, its values turned out to be close to 11 m/s, which is substantially lower than those established in other studies; this is explained by significant differences in the medium being fractured and in the impact velocity.

Keywords: *striker penetration, granite fragments, load characteristics, parameters of the broken-off piece, crack propagation velocity.*

Introduction

The process of crack development is decisive in the destruction of rocks by impact, which is implemented in some crushing machines. The study of the regularities of crack initiation and propagation upon impact is of considerable interest both for the general development of the theory of rock fracture and for its practical application, as evidenced by the significant number of studies devoted to this issue [1–7]. At the same time, owing to the difficulty of experimentally determining the crack propagation velocity v_{cr} in rocks, the number of works on this problem is very limited. They have mainly been carried out on homogeneous materials such as silicate glass, rosin, organic glass and some other polymers, whose transparency makes it possible to use optical methods for determining v_{cr}, which are characterized by comparatively simple measurements [8–12].

Studies of crack development in rocks [13–15] were carried out with cracks initiated by blasting or by a high-intensity ram impact, which very substantially affects the propagation velocity of the fracture crack [13] and exceeds by more than an order of magnitude the impact velocity of the working tool of machines used for impact destruction of rocks. The present work is devoted to estimating the

propagation velocity v_{cr} of a fracture crack in granite, as a typical representative of a heterogeneous rock, at an impact velocity on the rock of ~1 m/s.

Experimental procedure and results

The value of v_{cr} was determined taking into account the experimentally established length l_{cr} of the fracture crack that arises in a granite fragment when a piece is broken off it as a result of impact, and the calculated duration t_m of the process:

$$v_{cr} = l_{cr} / t_m. \quad (1)$$

In establishing l_{cr}, we proceeded from the assumption that the condition for the destruction of a rock fragment is that the crack arising as a result of the impact reaches one of its free surfaces [16].

The studies were carried out on an experimental setup (Fig. 1) used earlier in work [17]. Striker 2, in the form of a cylindrical hydraulic breaker pick with a flat blunting face of various diameters d, made of hardened steel, was fixed in a vertical position in a massive ram 3 (M = 2.8 kg). Under the striker, at a distance H = 32 mm, rock fragments 1 were placed — cores of granite from the Vyborg massif with the composition: 30–35 % quartz, 25–30 % plagioclase (Fig. 2), 30–35 % potassium feldspar, 5–10 % biotite, with strength characteristics established according to GOST 30629-2011: hardness f = 13 on the M. M. Protodyakonov scale, ultimate compressive strength σ_{comp} ~ 190 MPa. The aggregate hardness of the fragments used, determined by the Brinell method and converted to Vickers hardness, was ~450 HV; the hardness of the structural constituents of the rock reached 1200 HV (for quartz).

[Figure 1 — see original PDF, page 5] Setup for studying the process of impact destruction of rock (a) and its schematic diagram (b): 1 — rock fragment; 2 — striker; 3 — ram; 4 — roller; 5 — cam; 6 — gear reducer; 7 — electric motor

[Figure 2 — see original PDF, page 5] Appearance of quartz (light) and plagioclase (grey) grains in the structure of the material of the tested fragments

After impacts were delivered at various points of the horizontal surface of fragments 1 (the kinetic energy A_0 and the pre-impact velocity v_0 of the ram with the striker, calculated from the values of M and H, were 1.16 J and 0.91 m/s), the maximum distance of the impact point from the edge of their lateral surface at which a piece of rock is still broken off was determined. The geometric parameters of the spall site were measured, and the value of l_{cr} was established from them.

In separate experiments, to confirm the validity of the model, the depth of the craters formed in the granite upon impact was measured.

As a result of the experiments, the following was established:

- as in the case of a wedge at an open face wall [18], there exists a maximum, most effective distance h = l_{eff} of the impact point (point A in Fig. 3) from the edge of the fragments, at which the impact causes a piece of rock of maximum volume to break off from their lateral surface. At l < l_{eff} all the pieces broken off have a smaller volume; at l > l_{eff} several impacts have to be delivered at the selected point to break off a piece;
- the value of l_{eff} depends on the diameter d of the blunting flat of the striker and decreases as it increases;

- the broken-off pieces of rock are figures with a base in the form of a segment of the shape shown in Fig. 3, close to that described in [18], with parameters a, b, c, h and angles α and γ;
- parameters a, b, c are approximately equal to each other (Table 1) and decrease as d increases;
- the values of angle α vary in the range 112–134°, are close to the value of this parameter established in [18] (~120°), and are related to angle γ by the approximate ratio $\alpha \approx 2\gamma$.

[Figure 3 — see original PDF, page 5] View of the spall site after impact showing its geometric parameters

Owing to the similarity of the shape of all the broken-off pieces, it was assumed that in all experiments the fracture crack propagates at an angle γ ~ 60° from the impact point to the lateral surface of the fragments, and that its length l_{cr} for each diameter d can be taken as the arithmetic mean of the parameters a, b, c of the corresponding spall site (Table 1).

Table 1. Value of l_{eff} and mean values (from the results of 10 impacts) of the geometric parameters of the broken-off piece for impacts delivered by a tool with different diameters of the blunting flat

d, mm	l_{eff}, mm	a, mm	b, mm	c, mm	α, deg.	γ, deg.
1.12	4.0	8.59	8.68	9.01	128.6	63.6
1.36	3.0	7.31	6.78	7.13	127.4	64.5
1.68	2.5	5.37	5.47	5.25	121.3	61.3

Model of striker penetration into rock. Results of calculating the crack propagation time

Since the striker together with the ram constituted a structure of comparatively small length, it was considered that the influence of wave processes on the impact parameters can be neglected and that the interaction between the striker and the material being fractured can be described by the equations of a classical two-element impact system [16].

Based on the conclusions about the existing types of load characteristics of fracture [16], and taking into account additional observations of the impact process, it was assumed that the penetration of the striker into granite under the experimental conditions is characterized by three main stages:

- elastic deformation of the rock until the stress σ_c is reached over its area of contact with the striker, at which explosion-like fracture occurs under the contact area;
- advance of the striker through the crushed core of the rock, with part of the products being displaced from the crater and the remaining products being compacted without reaching the bottom of the fracture zone;
- rebound of the striker under the forces of residual elasticity of the compacted fracture products.

[Figure 4 — see original PDF, page 6] Form of the load characteristic of the system adopted in calculating the impact parameters

The dependence of the rock resistance force N on the deformation α (Fig. 4) for this case consists of three force functions: loading N_1 during elastic deformation of the rock under the contact area, loading N_2 during penetration of the striker into the rock fracture products, and unloading N_3, each

of which is characterized by its own stiffness (g_1, g_2, g_3). It was assumed that the functions N_1, N_2, N_3, as for the penetration of a body with a constant contact area, are linear. The area of the figure in Fig. 4 is the loading work A_L; the hatched area is the work A_f expended on fracture.

The maximum rock resistance force N_m and penetration α_p, as well as the impact time t_m for such a load characteristic, are [16]:

$$N_m = N_f \cdot \sqrt{[\ 1 - g_2/g_1 + (2A_L + g_2)/N_f^2\ ]}, (2)$$

$$\alpha_p = \alpha_m - N_m/g_3 = N_f \cdot (1/g_2 - 1/g_1) \cdot (\ \sqrt{[\ 1 - g_2/g_1 + (2A_L + g_2)/N_f^2\ ]} - 1\ ), (3)$$

$$t_m = \sqrt{(M/g_2)} \cdot arctg\ \sqrt{[\ (2A_L + g_2)/N_f^2 - g_2/g_1\ ]}, (4)$$

where M is the mass of the falling object (ram and striker); α_m is the maximum penetration of the striker into the rock; $g_1 = 2E_r d / [3(1 - \mu_r^2)]$; N_f is the contact fracture force of the rock, $N_f = f_c\sigma_{crit}$, where $f_c = \pi d^2/4$ is the contact area, which depends on the blunting flat of the striker; σ_{crit} is the stress at the contact point at which the tensile and shear stresses arising in the rock upon impact lead to its explosion-like fracture under the contact area.

[Figure 5 — see original PDF, page 6] Calculated parameters of a single impact t_m (1), N_m (2), σ_c (3), α_p (4) as functions of the diameter d of the blunting flat of the striker, compared with the experimentally established depth of the craters obtained

Assumptions adopted in calculating the impact duration:

- the critical contact stress σ_{crit} of granite can be taken equal to its dynamic hardness σ_{dyn} ~1300 MPa [19]. (Such a value of σ_{dyn}, compared with σ_{comp} = 190 MPa for granite, agrees with the conclusions of [20] that under uniaxial compression most rocks increase their strength characteristics 5–10 times when passing from static (σ_{comp}) to dynamic (σ_{dyn}) loading);
- by analogy with the stiffness ratios g_2/g_1 and g_3/g_1, which can be established from an analysis of the experimental force characteristics of contact fracture of brittle rocks given in [16], the stiffness g_2 in our case can be taken equal to $0.15g_1$, and g_3 equal to g_1;
- only about half of the kinetic energy A_0 of the striker is spent on elastic deformation and fracture of the rock (58.9 % for granite [19]); the rest of the energy goes to heat losses to the environment, elastic deformation of the striker and wear of its contact surface, as a result of which it was assumed that $A_f = 0.589 \cdot A_0 = 0.68$ J.

The numerical determination of the dependences of N_m, α_p, t_m, σ_c on the diameter d of the blunting flat of the striker was carried out in Microsoft Excel by substituting the parameters of the impact and of the rock being fractured into the derived equations (2)–(4).

The graph in Fig. 5 shows the values of the impact time t_m (curve 1), the rock resistance force N_m (curve 2), the contact stress $\sigma_c = N_m/(\pi d^2/4)$ (curve 3) and the penetration α_p of the striker into the rock (curve 4), calculated from equations (2)–(4) using the value of N_f for various diameters d of the contact area. The same graph shows, for the corresponding values of d, the crater depths established experimentally.

As can be seen from the graph in Fig. 5, at a constant force impulse $Mv_0 = N_m t_m$, as the diameter d of the contact area (blunting flat of the striker) increases, the maximum rock resistance force N_m increases monotonically, while the impact time t_m, the maximum contact stress σ_c and the penetration depth α_p of the indenter decrease. At the same time, as can be concluded from a comparison of the positions of the experimental points and curve 4, the regression line of the dependence $\alpha_p = f(d)$ shows satisfactory agreement with the experimental data, which confirms the validity of the proposed mathematical model and of the assumptions made.

Discussion of results. Conclusions

The propagation time of the fracture crack when a piece is broken off rock fragments by a striker with different diameters d of the blunting flat, taken as the impact time t_m of curve 1 (Fig. 5), is presented in Table 2. The same table gives the values of l_{cr}, as the arithmetic mean of the parameters a, b, c of the spall site (Table 1), and the results of calculating v_{cr} by formula (1).

Table 2. Parameters used to establish the propagation velocity of the fracture crack in granite fragments for strikers with different blunting diameters, and the values of v_{cr} obtained

d, mm	l_{cr}, mm	t_m, ms	v_{cr}^*, m/s
1.12	9.0	0.79	11.4
1.36	7.0	0.61	11.5
1.68	5.2	0.49	10.7

* Mean value v_{cr} = 11.2 m/s.

As can be seen from the table, the established propagation velocity v_{cr} of the fracture crack in the granite fragments, regardless of the size of the blunting flat of the striker and the impact duration, turned out to be close to 11 m/s.

Regarding the established value of v_{cr}, it may be noted that it differs substantially (being lower) from the values found in the literature, which range from 150 [11] to 2700 [15] m/s. Such a substantial difference may have the following explanation.

The data given in [11], which are usually cited by other authors, for example [21], were obtained by the polarization-optical method using high-speed filming. As the object modelling the rocks, samples of an optically active material, epoxymal, were used — a cross-linked polymer based on epoxy resin with maleic anhydride as the hardener. Clearly, the crack propagation velocity in such an isotropic medium, free of foreign inclusions, will differ substantially (being higher) from the velocity in granite, which consists of mineral grains and pore space whose share in the total rock volume reaches 1.01 % (for granites from the deposits of Karelia used in the present experiments to make the fragments to be fractured), and for some of its constituents (microcline) 3.28 % [22, 23]. The impossibility of quantitatively studying the fracture processes of crystalline rocks using amorphous plastics is also pointed out by the authors of [11] themselves.

The rock fracture velocities established in [15] were obtained by oscillographic recording of the moments at which conductive layers applied to the surface of the rock sample being fractured were broken by the moving crack. The experiments were carried out at ram impact velocities on the rock several orders of magnitude higher than the impact velocity in the experiments presented here. The

authors of [15] note that the crack development velocity in rocks depends on the impact velocity and decreases as it decreases, in accordance with the data given in Table 3.

Table 3. Dependence of the crack development velocity v_{cr} on the ram impact velocity on the rock v_{imp} [15]

Rock	v_{cr}, m/s, at impact velocity v_{imp} (m/s)			
	1200	830	500	350
Serpentinite	1140	1180	815	570
Limestone	2700	2150	850	790
Shale	2400	2250	1350	810

Extrapolation of the dependences in Table 3 to the level of impact velocities that occurred in our experiments (~1 m/s) gives values of v_{cr} of the same order as obtained in the present study.

References

1. Городилов Л. В., Ефимов В. П. Экспериментальное исследование трещинообразования при ударном разрушении породного массива твердым индентором // Фундаментальные и прикладные вопросы горных наук. 2018. Т. 5, № 1. С. 25–30.
2. Li Sh., Zhao P., Du X., Xiao P. Experimental study on crack evolution characteristics of rock-like materials under different strain rates // Journal of Geophysics and Engineering. 2018. Vol. 15, Iss. 5. P. 2071–2078.
3. Liu Y., Xu J., Zhou G. Relation between crack propagation and internal damage in sandstone during shear failure // Journal of Geophysics and Engineering. 2018. Vol. 15, Iss. 5. P. 2104–2109.
4. Wang X., Wu S., Ge H., Sun Y., Zhang Q. The complexity of the fracture network in the failure rock under cyclic loading and its characteristics of acoustic emission monitoring // Journal of Geophysics and Engineering. 2018. Vol. 15, Iss. 5. P. 2091–2103.
5. Xu J., Zheng Z., Xiao X., Li Z. Crack propagation and coalescence due to dual non-penetrating surface flaws and their effect on the strength of rock-like material // Journal of Geophysics and Engineering. 2018. Vol. 15, Iss. 3. P. 938–951.
6. Габов В. В., Задков Д. А., Нгуен Кхак Линь. Особенности формирования элементарных сколов в процессе резания углей и изотропных материалов эталонным резцом горных машин // Записки Горного института. 2019. Т. 236. С. 153–161.
7. Халкечев Р. К., Халкечев К. В. Управление селективностью разрушения горных пород на основе методов подобия и размерности в динамике трещин // Горный журнал. 2016. № 6. С. 64–66. DOI: 10.17580/gzh.2016.06.04.
8. Lundborg N., Johansson C. H. Experimental determination of the speed of propagation of cracks in glass as a function of the stress // Arkiv for Fysik. 1952. Vol. 4, No. 39. P. 555–558.
9. Кузьмин Е. А., Пух В. П. Скорость роста хрупкой трещины в стекле и канифоли // Некоторые проблемы прочности твердого тела: сб. статей. М.–Л.: Изд-во АН СССР, 1959. С. 367–374.
10. Регель В. Р. К вопросу о кинетике роста трещин в процессе разрушения твердых тел // Журнал технической физики. 1956. Т. 26, № 2. С. 359–363.
11. Бабенков И. С. Исследование взаимодействия бурового инструмента и породы методом фотоупругости. М.: Недра, 1970. 128 с.

12. Aman S. Microwave based method of monitoring crack formation // Measurement Science and Technology. 2014. Vol. 25, Iss. 5. P. 1–5.
13. Гаек Ю. В., Друкованный М. Ф., Мишин В. В. О скорости развития трещин в горных породах и твердых телах и методы ее измерения // Взрывное дело. 1963. № 51/8. С. 85–96.
14. Данчев П. С., Пучков Я. М., Ветлужских В. П. О скорости распространения трещин в прочной среде при взрыве // Вопросы разрушения горных пород взрывом: Тр. ИГД УФ АН СССР. Вып. 7. Свердловск: УФ АН СССР, 1963. С. 5–13.
15. Лагунов В. А., Мамбетов Ш. А. Скорость развития трещин в образцах горных пород // Прикладная механика и техническая физика. 1965. № 6. С. 96–99.
16. Соколинский В. Б. Машины ударного разрушения. М.: Машиностроение, 1982. 185 с.
17. Болобов В. И., Ле Тхань Бинь Закономерности разрушения материала ударника при повторяющихся единичных ударах // Записки Горного института. 2018. Т. 233. С. 508–512.
18. Зеленин А. Н. Основы разрушения грунтов механическими способами. М.: Машиностроение. 1968. 376 с.
19. Шелковников И. Г. Использование энергии удара в процессах бурения. М.: Недра, 1977. 160 с.
20. Ставригин А. Н., Тарасов Б. Г. Экспериментальная физика и механика горных пород. СПб.: Наука, 2001. 343 с.
21. Ушаков Л. С., Котылев Ю. Е., Кравченко В. А. Гидравлические машины ударного действия. М.: Машиностроение, 2000. 416 с.
22. Вайсберг Л. А., Каменева Е. Е. Возможности компьютерной рентгеновской микротомографии при исследовании физико-механических свойств горных пород // Горный журнал. 2014. № 9. С. 85–90.
23. Вайсберг Л. А., Каменева Е. Е. Взаимосвязь структурных особенностей и физико-механических свойств горных пород // Горный журнал. 2017. № 9. С. 53–58. DOI: 10.17580/gzh.2017.09.10.